

Bachelor of Science — Chemical Sciences

Table 1: Chemical Sciences — B23 Credit Distribution

Division	Subdivision	Credits
IC Core Courses	IC Compulsory	31
	IC Basket	9
	IKS + HSS	12
Discipline Courses	Discipline Core	62
	Discipline Electives	20
Electives	Free Electives	15
	Research Lit Presentation (1+1)	2
	RP (equivalent as DE)	12
Total		163

Table 2: B. S. Chemical Sciences — B23 Discipline Core Courses

Sl. No	Course Code	Course Name	L-T-P-C	Credits
1	CY-201P	Physical Chemistry Laboratory	0-0-4-2	2
2	CY-301	Principles and Theories of Physical Chemistry	3-0-0-3	3
3	CY-302	Principles of Organic Chemistry	3-0-0-3	3
4	CY-303	Fundamentals of Inorganic Chemistry	3-0-0-3	3
5	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
6	CY-202P	Organic Chemistry Laboratory	0-0-4-2	2
7	CY-203P	Inorganic Chemistry Laboratory	0-0-4-2	2
8	CY-304	Fundamental Analytical Chemistry	3-0-0-3	3
9	CY-401	Introduction to Quantum Chemistry and Molecular Spectroscopy	3-0-0-3	3
10	CY-512	Advanced Quantum Chemistry	3-0-0-3	3
11	CY-512P	Physical Chemistry Laboratory		3
12	CY-531	Organic Reactions and Mechanisms		3
13	CY-533	Chemistry of Main Group Elements		3
14	CY-533P	Inorganic Chemistry Laboratory		3
15	CY-511	Group Theory and Spectroscopy	3-0-0-3	3
16	CY-531P	Organic Chemistry Laboratory		3
17	CY-532	Photochemistry and Pericyclic Reactions		3
18	CY-534	Chemistry of Transition Elements		3
19	CY-514	Chemical and Statistical Thermodynamics	3-0-0-3	3
20	CY-535	Introduction to Organometallic Chemistry		3
21	CY-504	Heterocyclic Chemistry	2-0-0-2	2
22	CY-513	Chemical Kinetics and Reaction Dynamics	3-0-0-3	3
Total				62

Table 3: B. S. Chemical Sciences — B23 Semester 1

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-102P	Foundations of Design Practicum	1-0-6-4	4
2	IC	IC-112	Calculus	1.5-0.5-0-2	2
3	IC	IC-113	Complex Variables and Vector Calculus	1.5-0.5-0-2	2
4	IC	IC-152	Introduction to Python and Data Science	3-0-2-4	4
5	IC Basket	IC-131	Applied Chemistry for Engineers	2-0-2-3	3
6	IKS + HSS (choose 1)	IC-181	Indian Knowledge Systems	2-0-2-3	3
7		HS-342	German I	3-0-0-3	
8		HS-108	Basic English for Engineers	3-0-0-3	
Total					

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 4: B. S. Chemical Sciences — B23 Semester 2

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	IC	IC-114	Linear Algebra	2-0-0-2	2
2	IC	IC-115	ODE & Integral Transforms	2-0-0-2	2
3	IC	IC-140	Engineering Graphics for Design	2-0-3-4	4
4	IC	IC-161	Applied Electronics	3	3
5	IC	IC-161P	Applied Electronics Lab	2	2
6	IC	IC-222P	Physics Practicum	0-0-3-2	2
7	IC	IC-252	Probability and Statistics	3-0-2-4	4
8	IC Basket	IC-121	Mechanics of Particles and Waves	2.5-0.5-0-3	3
9	IKS + HSS (choose 0-3 credits)	IC-181	Indian Knowledge Systems	2-0-2-3	0-3
10		HS-151	Introduction to English Literature	3-0-0-3	
11		HS-357	French Language and Culture	2-0-0-2	
12		HS-308	European Literature	2-0-0-2	
Total					22-25

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 5: B. S. Chemical Sciences — B23 Semester 3

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-201P	Physical Chemistry Laboratory	0-0-4-2	2
2	DC	CY-301	Principles and Theories of Physical Chemistry	3-0-0-3	3
3	DC	CY-302	Principles of Organic Chemistry	3-0-0-3	3
4	DC	CY-303	Fundamentals of Inorganic Chemistry	3-0-0-3	3
5	DC	IC-136	Understanding Biotechnology and its Applications	3-0-0-3	3
Total					14

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 6: B. S. Chemical Sciences — B23 Semester 4

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-202P	Organic Chemistry Laboratory	0-0-4-2	2
2	DC	CY-203P	Inorganic Chemistry Laboratory	0-0-4-2	2
3	DC	CY-304	Fundamental Analytical Chemistry	3-0-0-3	3
4	DC	CY-401	Introduction to Quantum Chemistry and Molecular Spectroscopy	3-0-0-3	3
Total					10

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 7: B. S. Chemical Sciences — B23 Semester 5

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-512	Advanced Quantum Chemistry	3-0-0-3	3
2	DC	CY-512P	Physical Chemistry Laboratory		3
3	DC	CY-531	Organic Reactions and Mechanisms		3
4	DC	CY-533	Chemistry of Main Group Elements		3
5	DC	CY-533P	Inorganic Chemistry Laboratory		3
6	RP	DP-403P	Research Literature Presentation 1		1
Total					16

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 8: B. S. Chemical Sciences — B23 Semester 6

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-511	Group Theory and Spectroscopy	3-0-0-3	3
2	DC	CY-531P	Organic Chemistry Laboratory		3
3	DC	CY-532	Photochemistry and Pericyclic Reactions		3
4	DC	CY-534	Chemistry of Transition Elements		3
5	RP	DP-404P	Research Literature Presentation 2		1
Total					13

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 9: B. S. Chemical Sciences — B23 Semester 7

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-514	Chemical and Statistical Thermodynamics	3-0-0-3	3
2	DC	CY-535	Introduction to Organometallic Chemistry		3
3	RP	DP-551P	Undergraduate Research Project 1		6
Total					12

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

Table 10: B. S. Chemical Sciences — B23 Semester 8

Sl. No	Course Category	Course Code	Course Name	L-T-P-C	Credits
1	DC	CY-504	Heterocyclic Chemistry	2-0-0-2	2
2	DC	CY-513	Chemical Kinetics and Reaction Dynamics	3-0-0-3	3
3	RP	DP-552P	Undergraduate Research Project 2		6
Total					11

Aim for 20–25 credits this semester — top up the fixed core above with Discipline / Free Electives so the elective budget is covered on time.

*Beyond the core courses above, students choose 20 credits of **Discipline Electives** and 15 credits of **Free Electives** from the approved lists (choice is the student's; see the Available Discipline Electives table below).*

Table 11: Chemical Sciences — Available Discipline Electives (choose from these)

Sl. No	Course Code	Course Name	Credits
1	BY-505	Nanobiotechnology	3
2	BY-514	Analytical Biotechniques	3
3	CE-519	Chemistry of Natural Waters	3
4	CH-101	General Chemistry	4
5	CY-001	Preparatory Chemistry - 1	4
6	CY-002	Preparatory Chemistry - 2	4
7	CY-100	Preparatory Chemistry 1	3
8	CY-101	Chemistry	5
9	CY-200	Foundations and applications of Chemistry	3
10	CY-241	Nano-scale Science and Technology	3
11	CY-242	Introduction to Molecular Spectroscopy	3
12	CY-243	Engineering Chemistry	3
13	CY-247	Introduction to Molecular Thermodynamics	3
14	CY-248	Molecular Physical Chemistry for Engineers	3
15	CY-342	Understanding Small Systems	3
16	CY-344	Food Chemistry: Processing, Preservation and Storage	3
17	CY-402	Applied Materials Chemistry	3
18	CY-403	Numerical Methods and Data Analysis in Chemistry	3
19	CY-404	Fundamentals of Soft Matter Science and Applications	3
20	CY-405	Introduction to structural chemistry	3
21	CY-501	Organic Reaction and Mechanism	4
22	CY-501P	Organic Chemistry Laboratory	4
23	CY-502	Photochemistry and pericyclic reactions	4
24	CY-506	Chemistry of Transition Elements	4
25	CY-506P	Inorganic Chemistry Lab	4
26	CY-507	Chemistry of Main Group Elements	4
27	CY-508	Organometallic Chemistry	2
28	CY-511P	Physical Chemistry Laboratory	4
29	CY-515	Advanced Inorganic Spectroscopy	3
30	CY-521	Mathematics for Chemists	3
31	CY-522	Computational Chemistry	3
32	CY-523	Colloids and Interface Science and Technology	3
33	CY-524	Basic and Applied Electrochemistry	3
34	CY-541	Fundamentals of Organic Chemistry	3
35	CY-545	Reagents in organic synthesis	3
36	CY-547	Chemical Crystallography	3
37	CY-550	Bioinspired Materials	3
38	CY-552	Hydrogen Generation and Storage	3
39	CY-553	Organic Inorganic Spectroscopy	3
40	CY-554	Science and Technology of Nanomaterials	3

Chemical Sciences — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
41	CY-555	Introduction to Polymer Science & Technology	3
42	CY-556	Organic Spectroscopy	3
43	CY-560	Introduction to Supramolecular Chemistry	3
44	CY-561	Natural Products Synthesis	3
45	CY-562	Stereochemistry of organic compounds	3
46	CY-562P	Design Practicum - II	1
47	CY-598P	Post Graduate Project-1 (CH)	4
48	CY-599P	Post Graduate Project-2	9
49	CY-641	Polymer Synthesis	3
50	CY-642	Molecular- and Bio-electronics	3
51	CY-643	Advanced Analytical Techniques	3
52	CY-644	Bioinorganic chemistry	3
53	CY-645	Reagents in organic synthesis	3
54	CY-646	Advanced NMR Spectroscopy - A Problem Based Approach	3
55	CY-660	Photocatalysis: Concepts and Applications	3
56	CY-670	Fluorescence spectroscopy, microscopy and applications	3
57	CY-701	Advances Physical Methods in Chemistry Theory and	3
58	CY-702	Advanced Inorganic Chemistry: Theory and Applications	3
59	CY-703	Advanced Organic Chemistry	3
60	CY-704	Introduction to Theoretical Chemistry	3
61	CY-705	Modern Methods in Organic Synthesis	3
62	CY-746	Self Assembly of Surfactants and Polymers in Solution	3
63	EN-503	Energy Storage Technologies	3
64	ME-353	Electronic Materials and Their Applications	3
65	ME-509	Nanomanufacturing	3
66	ME-515	Carbon Materials and Technology	3
67	ME-516	Polymer Technology for Engineers	3
68	MT-201	Physics of Solids	3
69	MT-501	Energy Conversion and Storage Technologies	3
70	MT-502	Recycling and Circular Economy	3
71	MT-503	Semiconductor Materials and Devices	3
72	MT-505	Thin Film Technology	3
73	MT-506	Biomaterials	3
74	MT-509	Hydrogen Energy	3
75	MT-510	Colloids and Interfaces	3
76	PH-504	Organic Optoelectronics	3
<i>Compulsory project components — counted as DE, not selectable</i>			
77	DP-551P	Undergraduate Research Project 1 (compulsory; counted as DE)	6

Chemical Sciences — Available Discipline Electives (choose from these) (continued)

Sl. No	Course Code	Course Name	Credits
78	DP-552P	Undergraduate Research Project 2 (compulsory; counted as DE)	6